



kubectl get pods in Kubernetes

Introduction

`kubectl get pods` is one of the most commonly used **Kubernetes commands** to list and monitor running Pods in a cluster.

- What does `kubectl get pods` do?
- Why is it important?
- How can we use different options to get useful details?

This guide will help you **understand** `kubectl get pods` with practical examples.

Section 1: Learn

What is `kubectl get pods`?

`kubectl get pods` is used to **list, filter, and monitor** all running and non-running Pods in a cluster.

Feature	Description
Lists All Pods	Displays all running and pending Pods
Shows Status	Helps check if a Pod is running, pending, or failed
Filters by Namespace	Allows viewing Pods in specific namespaces
Displays Wide Information	Provides additional details like node assignment



Why Use `kubectl get pods`?

1. **Quick cluster status check** – Know which Pods are running or failing.
2. **Monitor application health** – Helps identify problems with deployments.
3. **Troubleshoot issues** – Identify Pods in **CrashLoopBackOff** or **Error** states.
4. **Filter and sort** – View specific Pods based on names, labels, and status.

Interesting Fact

In Kubernetes, a Pod is the **smallest deployable unit**, and `kubectl get pods` is the **simplest way** to monitor its status.

Section 2: Practice

1. Basic Usage

List All Running Pods in the Default Namespace

```
kubectl get pods
```

List All Pods in a Specific Namespace

```
kubectl get pods -n kube-system
```

2. Detailed Output

Get Pods with More Details (Wide Format)

```
kubectl get pods -o wide
```



Check Specific Pod Information

```
kubectl get pod <pod-name> -o yaml
```

3. Filtering and Sorting

List Only Running Pods

```
kubectl get pods --field-selector=status.phase=Running
```

List Only Failed Pods

```
kubectl get pods --field-selector=status.phase=Failed
```

Get Pods Matching a Label

```
kubectl get pods -l app=my-app
```

4. Watching Pods in Real Time

Watch for Changes in Pod Status

```
kubectl get pods --watch
```

Get Live Updates of a Pod

```
kubectl get pods <pod-name> --watch
```

Section 3: Know More

Frequently Asked Questions

1. What does **kubectl get pods** show?



- It lists all running, pending, or failed Pods in the cluster.
 - 2. **How do I check why a Pod is not running?**
 - Use `kubectl describe pod <pod-name>` to see error details.
 - 3. **How do I check Pods in all namespaces?**
 - Use `kubectl get pods --all-namespaces`.
 - 4. **What does `kubectl get pods -o wide` do?**
 - It provides additional details such as node assignment and IP addresses.
 - 5. **How do I continuously monitor Pod status?**
 - Use `kubectl get pods --watch` for real-time updates.
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Conclusion

`kubectl get pods` is the **most essential command** for monitoring Kubernetes applications.

It provides **real-time insights** into Pod status, **helps in debugging**, and ensures **smooth cluster operations**.